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Date: 10/12/98, 4:31 PM
Re: Cold Fusion Revisited

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From: A.J. Craddock [REDACTED]
Subject: Cold Fusion Revisited
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Subject: Cold Fusion

Cold Fusion: The "Miracle" is No Mistake

(Excerpts from Analog Magazine

July / August 1997
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The acid test for any revolutionary discovery--entry to the marketplace--is happening right now for cold fusion, whether or not the critics agree. It is now possible to acquire for research purposes - a low-power, radiationless, tabletop nuclear reactor for less than \$4,000!

Hot fusion has never produced a single watt of excess energy, and if it ever does it would be in the form of lethal radiation. Cold fusion, whose primary signature is excess heat with no deadly radiation, exceeded breakeven at birth -- in 1989.

Already, former chairman of the House Science Committee, [REDACTED] (R-PA), has publicly acknowledged the reality, importance, and mystery of cold fusion.

Many other "establishments" -- including the planet's entire fossil fuel and fission power infrastructure -- are also in for rude awakenings. Even hydropower, solar, and windmill energy, which may still have their place in the coming New Energy Age, will be marginalized. It will be extremely difficult, even for contemporary "sustainable energy", to compete with compact, environmentally benign power systems that do not bow

to Earth's rotation, or the vagaries of weather. The electric power grid will wither away as consumers increasingly adopt self contained domestic power units based on cold fusion, or "new hydrogen energy," as the Japanese official program calls it. Air pollution will plummet, and the threat of global warming will dissipate.

While the cold fusion triumph is astonishingly good news for the environment, and the energy well-being of civilization, It is extremely bad news for business-as-usual, incremental science. Mainstream scientific publications have been loathe to recognize what has been happening these past eight years in laboratories from Los Alamos to Beijing.

Both the nuclear-scale energy production and nuclear products -- low-energy, alchemy-like transmutation of heavy elements in cold fusion -- have now been proved.

The energy release observed so far from cold fusion reactions is slow (fortunately!) but spectacular: thousands to millions of times beyond any conceivable chemical reaction in those systems. Tens, hundreds, and thousands of megajoules per mole of active material in the cells, for those who understand those terms. Moreover, there are no chemical changes in cold fusion systems that could begin to explain the energy release -- much less the nuclear reaction products found: transmuted heavy elements, helium-4, occasionally radioactive tritium, and an extremely faint neutron emission (the latter often nonexistent, or very near ambient background level).

To be sure, these are provocative conclusions about one of the most reviled, disputed anomalies in the history of science.

They will provide historians with a fascinating view of skepticism run amok. They are each good examples of Arthur C. Clarke's observation in 1963: "It is really quite amazing by what margins competent but conservative scientists and engineers can miss the mark, when they start with the preconceived idea that what they are investigating is impossible. When this happens, the most well-informed men become blinded by their prejudices and are unable to see what lies directly in front of them"

The Electric Power Research Institute (EPRI) final report on its \$10 million cold fusion research states: "This work confirms the claims of Fleischmann, Pons, and Hawkins of the production of excess heat in deuterium loaded palladium cathodes at levels too huge for chemical transformation... . Further, the excess energy exceeds that of known chemical processes by two or more orders of magnitude."

Drs. Fleischmann and Pons have been vindicated by EPRI, but in other ways too.

Leaving the hostility of the U.S. scientific establishment behind.

The European patent office has given notice to U.S.-based cold fusion company ENECO of Salt Lake City that the Pons and Fleischmann patent will be approved. Not so in the U.S., as the war rages in Washington against cold fusion. The Department of Energy continues massive funding for various hot fusion programs and fission power research, but not one cent has been allocated for cold fusion. Any research on cold fusion at national labs has been either "bootlegged" under duress, or paid for out of "director's discretionary funding."

At last report, Pons and Fleischmann are resident in France and are being funded by Toyota.

Regards


Web Administrator
CSETI